Customer Segmentation & Churn Pattern Analytics in European Banking

Research Paper: exploratory analysis, findings and recommendations

Prepared by: Suryapriya

Role: Data Analyst Intern, Unified Mentor Pvt. Ltd.

Dataset: European_Bank.csv (10,000 customers: France, Spain, Germany)

Tools used: Python (Pandas, SciPy, scikit-learn, Matplotlib), Streamlit, Plotly

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Abstract

Banks usually know their overall churn rate, but that single number does not tell them who is leaving or why. In this project I analysed 10,000 retail banking customers from France, Spain and Germany to find out which groups churn the most. I split customers by country, age, tenure, credit score, balance, gender, product count and activity, and compared churn rates across each split.

About 20.4% of customers left the bank. The risk is not spread evenly. Germany loses 32.4% of its customers compared with roughly 16% in France and Spain, and customers aged 46 to 60 churn at 51%. Women churn more than men, inactive members churn almost twice as much as active ones, and customers holding three or four products churn at very high rates. Tenure, credit card ownership and salary show little or no relationship with churn. High-balance customers churn slightly more than average, and the churned high-value group holds about €88.7M, close to half of all the balances that left.

The results are turned into a Streamlit dashboard and a set of practical retention recommendations.

1. Introduction

1.1 Background

Customer churn is one of the quieter costs in retail banking. When a customer leaves, the bank loses the lifetime value of that relationship, has to spend more to acquire a replacement, and sees its revenue become less predictable. Most banks track the churn rate, but a single percentage is not much use when you have to decide where to spend a retention budget.

1.2 Problem statement

The bank has detailed customer-level data, but without structured analysis its retention efforts end up generic and reactive. Three things are missing: a clear view of which customer groups are at the highest risk, an understanding of how churn differs by geography and demographics, and a measure of the financial profile of the customers who leave.

1.3 Objectives

- Measure the overall churn rate.
- Find how churn is distributed across customer segments.
- Compare churn across the three European markets.
- Look specifically at high-value customers and the money at risk.
- Check how engagement and tenure relate to churn, and use the findings to support planning and marketing decisions.

2. Data and Methodology

2.1 Dataset

The dataset has 10,000 rows, one per customer, and 14 columns: customer ID, surname, credit score, geography, gender, age, tenure, balance, number of products, credit card ownership, activity status, estimated salary, the target column Exited (1 = left the bank), and a Year column that holds the value 2025 for every row. Because every row has the same year, this is a snapshot of one period and I could not study how churn changes over time.

2.2 Validation and cleaning

- There were no missing values and no duplicate rows, and every CustomerId is unique.
- HasCrCard, IsActiveMember and Exited only contain 0 and 1, so the binary fields are consistent.
- The ranges look sensible: age 18 to 92, credit score 350 to 850, tenure 0 to 10 years, number of products 1 to 4.
- I dropped Surname because it has no analytical value, and kept CustomerId only as an identifier.
- 36.2% of customers have a balance of exactly zero. I treated this as a real customer group instead of an error, and gave it its own segment.

2.3 How the segments were built

The brief names the segment types but not the exact cut-offs, so I chose the ones below. They are easy to explain and give groups large enough to compare. Anyone can change them in the code if the bank uses different definitions.

Dimension	Groups	Notes
Geography	France, Spain, Germany	As given in the data
Age	<30, 30–45, 46–60, 60+	60+ is the smallest group (464 customers)
Tenure	New (0–2 yrs), Mid-term (3–6), Long-term (7+)	Roughly 25% / 40% / 35% of customers
Credit score	Low (<580), Medium (580–669), High (670+)	Follows common credit score ranges
Balance	Zero, Low, High	Low/High split at the median of non-zero balances
High-value	Top 25% by balance (≥ €127,644)	2,500 customers; used for revenue-risk analysis

2.4 Analysis steps

For each segment I calculated the churn rate (churned customers divided by customers in the segment) and the share of all churn that the segment accounts for. These two numbers answer different questions. A small group can have a very high churn rate but not matter much overall, while a big group with a moderate rate can account for most of the lost customers. I then compared the average profile of churned and retained customers, tested whether the differences were statistically meaningful with chi-square tests, and trained a random forest model on a 75/25 split to see which variables carry the most information about churn.

3. Results

3.1 Overall churn

Of the 10,000 customers, 2,037 left, which gives an overall churn rate of 20.37%. Roughly one customer in five has gone. Every segment rate below should be read against this baseline.

3.2 Churn by geography

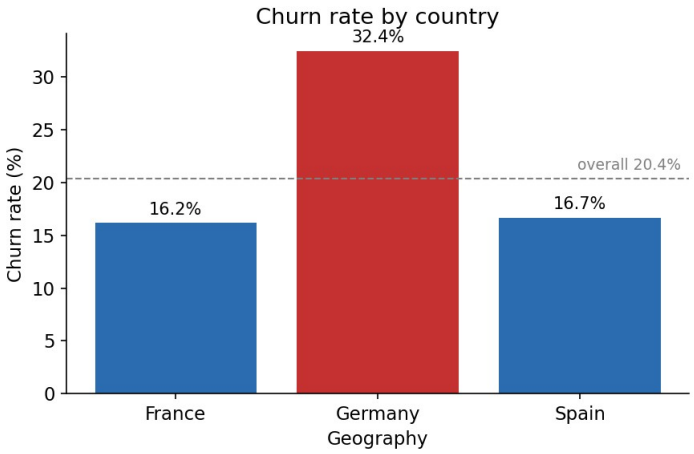


Figure 1. Churn rate by country

Country	Customers	Churned	Churn rate	Risk index
France	5,014	810	16.2%	0.79
Germany	2,509	814	32.4%	1.59
Spain	2,477	413	16.7%	0.82

Germany stands out. It has only a quarter of the customers but 40% of all churn, equal to France, which has twice as many customers. I defined the Geographic Risk Index as a country's churn rate divided by the overall rate. Germany scores 1.59, meaning its customers are about 60% more likely to leave than the average customer. France and Spain are both below 1.

The data cannot explain why Germany is different. It could be competition, pricing, product fit or service quality, and finding out would need information the dataset does not contain. What the data does show is that the German gap is not caused by one group. It appears in every age band and for both genders (see below), so it looks like a market-level issue.

3.3 Churn by age

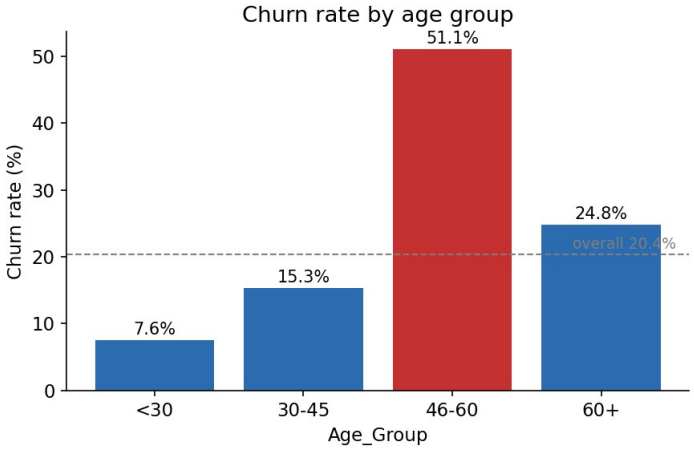


Figure 2. Churn rate by age group

Age is the strongest pattern in the whole dataset. Churn rises from 7.6% for customers under 30 to 15.3% for those aged 30–45, and then jumps to 51.1% for customers aged 46–60. After 60 it drops back to 24.8%. The 46–60 group is only 16.5% of customers but produces 41% of all churn.

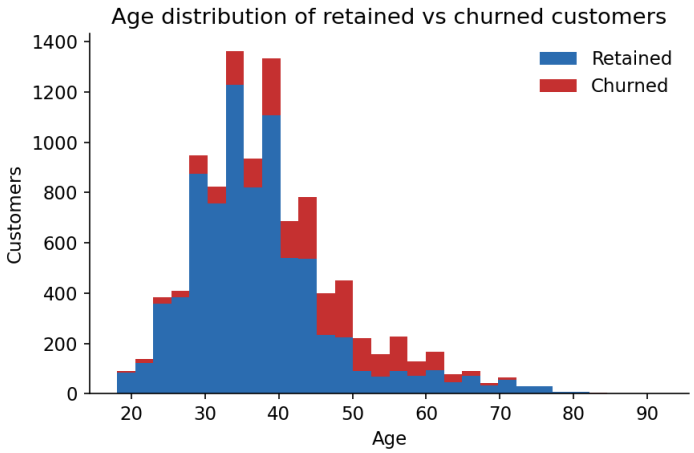


Figure 3. Age distribution of retained and churned customers

The average churned customer is 44.8 years old compared with 37.4 for retained customers, which matches this picture. A bank that treats a 25-year-old and a 52-year-old as the same kind of customer is missing the point.

3.4 Geography and age together

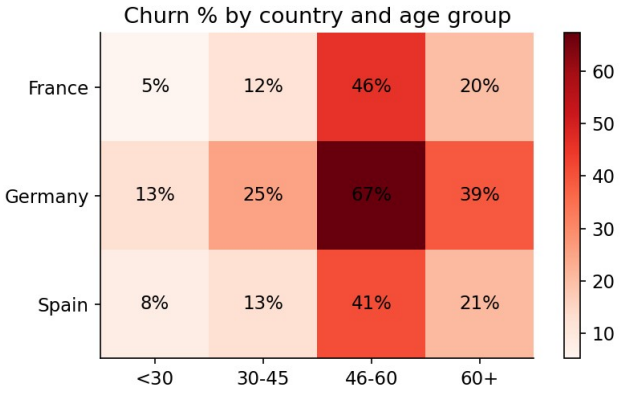


Figure 4. Churn % by country and age group

Putting the two together shows that the problems add up. In Germany, 67.3% of customers aged 46–60 left. The same age group churns at 45.8% in France and 40.7% in Spain, which is already high, but Germany is far above both. Even young German customers (under 30) churn at 12.6%, more than France (5.2%) or Spain (7.9%). So the 46–60 spike appears in every country, and Germany is simply worse at every age.

3.5 Gender

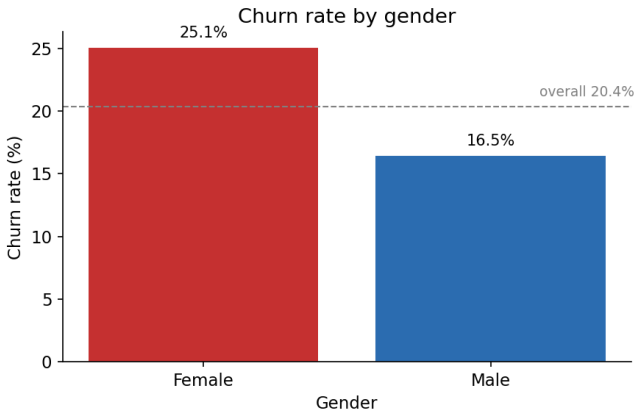


Figure 5. Churn rate by gender

Women churn at 25.1% and men at 16.5%. Women are 45% of customers but 56% of those who left. The gap holds in all three countries: in Germany it is 37.6% against 27.8%, in France 20.3% against 12.7%, and in Spain 21.2% against 13.1%.

3.6 Products and engagement

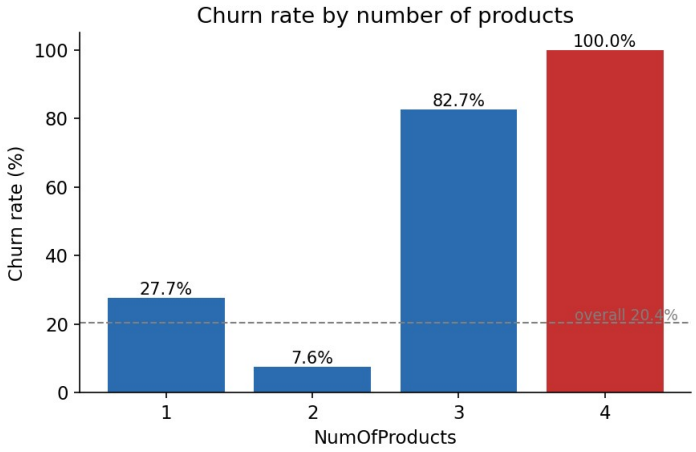


Figure 6. Churn rate by number of products

The number of products is closely tied to churn, but not in a straight line. Customers with one product churn at 27.7%, and those with two products churn at only 7.6%, by far the lowest rate. Then it reverses sharply: 82.7% of customers with three products left, and all 60 customers with four products left. The three and four product groups are small (326 customers combined), but a churn rate of around 86% should not be ignored. One possible reading is that these customers were pushed into extra products and were not satisfied, but the data cannot confirm this and it would be worth checking with the business.

Customers with only one product are the other concern. They are 51% of the customer base and account for 69% of all churn.

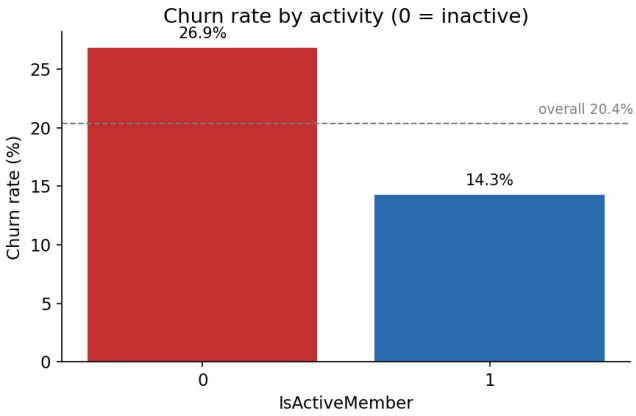


Figure 7. Churn rate by activity status

Activity matters as well. Inactive members churn at 26.9% against 14.3% for active members, a ratio of 1.88, which is the Engagement Drop Indicator in this project. Inactive customers are 48.5% of the base but 64% of all churn. The effect holds within product groups too: for example, among one-product customers, inactive ones churn at 36.7% and active ones at 18.9%.

3.7 Tenure, credit score and balance

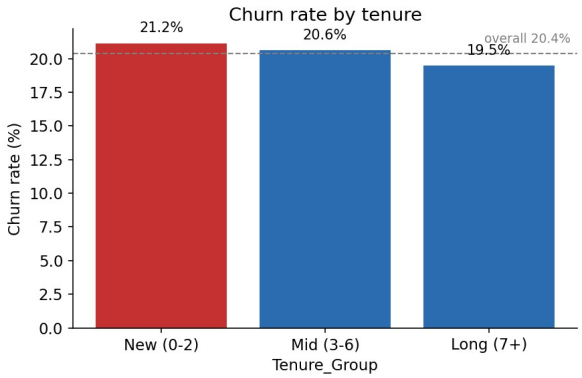


Figure 8. Churn rate by tenure

Tenure makes almost no difference. New customers churn at 21.2%, mid-term at 20.6% and long-term at 19.5%, and the chi-square test says this is not a significant difference (p = 0.25). This surprised me, because it is common to assume that loyal long-term customers are safer. In this data, they are not noticeably safer.

Credit score has only a small effect: 22.0% for the low band, 20.6% for medium and 19.3% for high. Credit card ownership has no significant relationship with churn at all (p = 0.49).

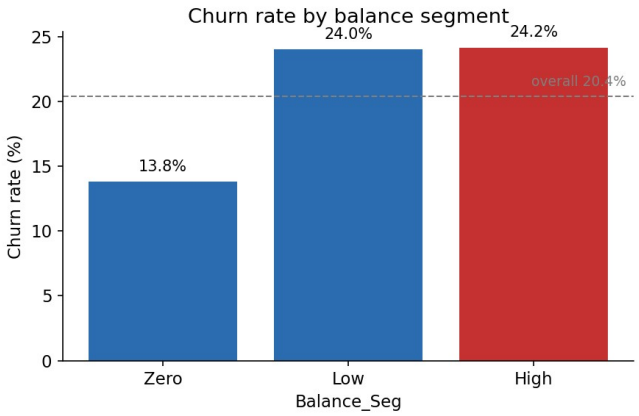


Figure 9. Churn rate by balance segment

Balance behaves in an interesting way. Customers with a zero balance churn the least (13.8%), while customers with a low or high balance both churn at about 24%. So having money in the account does not make a customer stay. If anything, customers who hold money with the bank are more likely to leave than those who do not.

3.8 Churned versus retained customers

Average of	Retained	Churned	Difference
Age	37.4	44.8	+7.4 yrs
Balance	€72,745	€91,109	+€18,364
Credit score	651.9	645.4	−6.5
Tenure (years)	5.0	4.9	−0.1
Estimated salary	€99,738	€101,466	+€1,728
Active member (share)	0.6	0.4	lower

The two groups look alike on tenure, salary and credit score. They differ in age, balance and activity. Salary in particular has almost no relationship with churn (correlation of 0.01), so a customer's income level tells us very little about whether they will leave.

3.9 High-value customer churn

I defined high-value customers as the top 25% by balance, which means a balance of €127,644 or more. That gives 2,500 customers, of whom 592 churned, a high-value churn ratio of 23.7%. For the remaining customers the rate is 19.3%, so premium customers leave a bit more often than everyone else, not less.

The financial side is where it matters. The customers who left held €185.6M in total, and the high-value churners alone held €88.7M of that, or 47.8%. Less than a quarter of customers produce nearly half of the balance that walked out of the door. I want to be clear that balance is a proxy for value. The dataset has no revenue or profit figures, so I cannot say how much income this money generated.

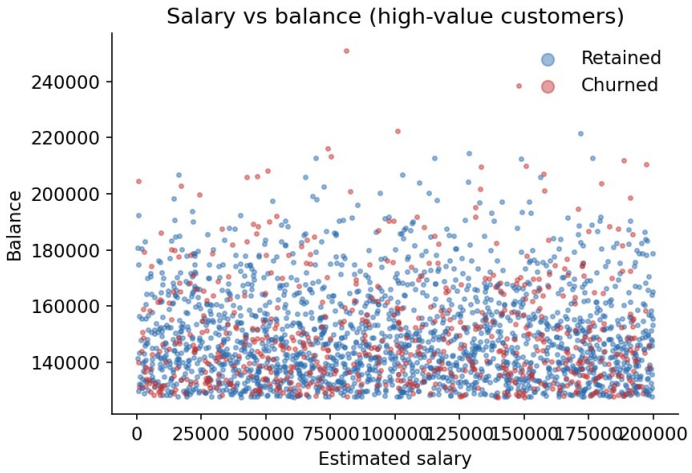


Figure 10. Salary vs balance for high-value customers

The scatter plot shows no clear salary pattern, since churned and retained customers are mixed across all salary levels. Inside the high-value group the same drivers show up as elsewhere: Germany has a high-value churn rate of 29.0% against about 20–21% in France and Spain, and inactive high-value customers churn at 30.5% compared with 16.9% for active ones.

3.10 Statistical tests and key drivers

Chi-square tests confirm that geography, gender, age group, balance segment, number of products and activity status are all significantly related to churn ($p < 0.001$). Tenure ($p = 0.25$) and credit card ownership ($p = 0.49$) are not, and credit score band is only borderline ($p = 0.03$).

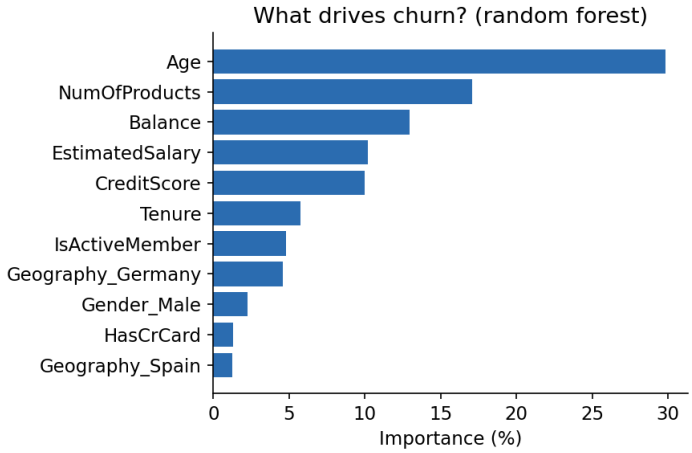


Figure 11. Relative importance of variables in a random forest model

A random forest model trained on 75% of the data scored an accuracy of 85.2% and an AUC of 0.87 on the remaining 25%. Age (about 30%), number of products (17%) and balance (13%) came out as the top three. I should note that salary and credit score also get some importance (around 10% each) mostly because continuous variables give the model many places to split. The earlier segment analysis shows they have almost no real relationship with churn, so I would not read that as evidence that they drive it. The model is used here only to confirm the ranking, not as a final prediction system.

4. Key Performance Indicators

KPI	What it measures	Value
Overall Churn Rate	% of customers who exited	20.4%
Segment Churn Rate	Churn % within a segment	See Section 3 (range 7.6% to 100%)
High-Value Churn Ratio	Churn among top 25% by balance	23.7%
Geographic Risk Index	Country churn rate ÷ overall rate	Germany 1.59, Spain 0.82, France 0.79
Engagement Drop Indicator	Inactive churn rate ÷ active churn rate	1.88 (26.9% vs 14.3%)

5. The Dashboard

All the numbers above are available in an interactive Streamlit dashboard (app.py in the project repository). The sidebar has filters for country, gender, age group, tenure, credit band and balance segment, and the KPIs at the top change whenever a filter is changed. There are five tabs: an overview of churned versus retained customers, a geography view with the risk index and a country-by-age heatmap, an age and tenure comparison, a high-value customer explorer where the balance threshold can be adjusted, and a drill-down tab to break any segment down by a second dimension. The filtered data can also be downloaded as a CSV file.

The code and the Colab notebook are on GitHub, so the dashboard can be run locally or in Google Colab.

6. Recommendations

These follow from the numbers above. They are suggestions for the bank to test, since the data shows association and not cause.

- **Treat Germany as a separate retention project.** It has the highest churn in every age and gender group. Before spending money, the bank should find out why, for example through exit surveys, a look at German competitors and a review of product pricing.
- **Build a retention programme for customers aged 46–60.** This one age band is 16.5% of customers and 41% of churn. German customers in this band (67% churn) should be the first group to contact.
- **Move single-product customers to a second product.** Customers with two products churn at 7.6% against 27.7% for one product. This may partly reflect customers who are already committed, so it is worth running as a small pilot before scaling up.
- **Look into the three and four product customers.** Their churn of about 86% is a warning sign. The bank should check whether these customers were sold products that did not suit them.
- **Re-engage inactive members, especially those with large balances.** Inactive high-value customers churn at 30.5%. A simple outreach call or personal offer for them is a cheap first step compared with the €88.7M exposure.
- **Create a monthly watch list.** Customers who are older, hold a single product, are inactive and are in Germany could be flagged automatically so that relationship managers can contact them before they leave.

- **Do not build campaigns around tenure, credit card ownership or salary.** None of them separates churners from non-churners in this data, so money spent there is unlikely to help.

7. Limitations

- The dataset is a single snapshot (all rows are from 2025). I cannot see trends over time or tell when in the year customers left.
- There are no revenue or profit figures. Balance is used as a stand-in for customer value, which is only an approximation.
- The segment cut-offs (for example credit bands and tenure groups) are my own choices. Different cut-offs could shift some percentages slightly, though the main patterns are strong enough that I do not expect them to change.
- The findings show association, not cause. For example, a low churn rate for two-product customers does not prove that selling a second product reduces churn.
- Several groups are small (60+ customers, 3 and 4 product holders), so their percentages are less stable than the rest.
- The data has no reasons for leaving, such as complaints, fees or service quality, which would be the most useful information for deciding on actions.

8. Conclusion

The overall churn rate of 20.4% hides large differences between customer groups. Churn is concentrated among German customers, people aged 46 to 60, women, inactive members and customers with one, three or four products. It is largely unrelated to tenure, credit card ownership and salary. High-balance customers are not safer than others, and the ones who left took about €88.7M with them.

This means a general retention campaign would waste effort. A better approach is to focus on the few segments where the risk is clearly highest, and to check the reasons behind those patterns with the business before acting. The dashboard built in this project lets analysts and managers explore those segments themselves and update the numbers as new data arrives.

If I continued this work, the next steps would be adding revenue data to measure value more accurately, collecting data over several periods to see trends, and building a proper churn prediction model with the early-warning list described above.